MANIPAL SCHOOL OF INFORMATION SCIENCES

Manipal Academy of Higher Education

M.Tech in Cyber Security

CRYPTOLOGY

Assignment

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1 Introduction

Cryptography is the technique of securing information by converting plaintext into ciphertext using an encryption algorithm and a key. Cryptanalysis is the process of studying and analyzing encrypted information to recover the original plaintext or determine the encryption key.

This assignment covers four important cryptographic concepts:

- Caesar cipher and ciphertext-only cryptanalysis.
- Affine cipher and chosen-plaintext attack.
- Affine cipher and known-plaintext attack.
- One-Time Pad encryption and decryption.

For all alphabet-based calculations, the following numerical representation is used:

$$A = 0,\ B = 1,\ C = 2,\ \ldots,\ Z = 25$$

All calculations are performed modulo 26.

2 Problem 1: Caesar Cipher Cryptanalysis

Problem Statement:

Caesar sends the ciphertext `EVIRE` to Mark Antony. Antony does not know the encryption key and attempts all possible keys to discover the meaningful plaintext.

2.1 Cryptanalysis Model

Model Used: Ciphertext-Only Attack

In a ciphertext-only attack, the attacker has access only to the encrypted message and does not know the encryption key or original plaintext.

The Caesar cipher has only 26 possible keys. Therefore, Antony can try every possible key through brute-force analysis.

2.2 Caesar Cipher Decryption Formula

The decryption formula is:

$$P = (C - K) \bmod 26$$

Where:

- P = Plaintext numerical value.
- C = Ciphertext numerical value.
- K = Secret key.

The ciphertext is:

`EVIRE`

2.3 Brute-Force Analysis

Antony tries all 26 possible keys from 0 to 25.

For example, when the key is 4:

$$E = 4, \quad V = 21, \quad I = 8, \quad R = 17, \quad E = 4$$

Applying the decryption formula:

$$P = (C - 4) \bmod 26$$

$$E : (4 - 4) \bmod 26 = 0 = A$$

$$V : (21 - 4) \bmod 26 = 17 = R$$

$$I : (8 - 4) \bmod 26 = 4 = E$$

$$R : (17 - 4) \bmod 26 = 13 = N$$

$$E : (4 - 4) \bmod 26 = 0 = A$$

Therefore:

$$\boxed{\texttt{ARENA}}$$

Similarly, when the key is 13:

$$P = (C - 13) \bmod 26$$

$$E : (4 - 13) \bmod 26 = 17 = R$$

$$V : (21 - 13) \bmod 26 = 8 = I$$

$$I : (8 - 13) \bmod 26 = 21 = V$$

$$R : (17 - 13) \bmod 26 = 4 = E$$

$$E : (4 - 13) \bmod 26 = 17 = R$$

Therefore:

$$\boxed{\texttt{RIVER}}$$

2.4 Results of All Possible Keys

The following table shows the plaintext obtained for every possible Caesar cipher key.

Table 1: Caesar Cipher Brute-Force Results

Key	Decrypted Plaintext	Observation
0	EVIRE	Not meaningful
1	DUHQD	Not meaningful
2	CTGPC	Not meaningful
3	BSFOB	Not meaningful
4	ARENA	Meaningful
5	ZQDMZ	Not meaningful
6	YPCLY	Not meaningful
7	XOBKX	Not meaningful
8	WNAJW	Not meaningful
9	VMZIV	Not meaningful
10	ULYHU	Not meaningful
11	TKXGT	Not meaningful
12	SJWFS	Not meaningful
13	RIVER	Meaningful
14	QHUDQ	Not meaningful
15	PGTCP	Not meaningful
16	OFSBO	Not meaningful
17	NERAN	Not meaningful
18	MDQZM	Not meaningful
19	LCPYL	Not meaningful
20	KBOXK	Not meaningful
21	JANWJ	Not meaningful
22	IZMVI	Not meaningful
23	HYLUH	Not meaningful
24	GXKTG	Not meaningful
25	FWJSF	Not meaningful

2.5 Answers

(a) Meaningful plaintexts:

The two meaningful plaintexts obtained are:

`ARENA`

RIVER

(b) Meeting location:

The word RIVER suggests the Tiber River, while ARENA suggests the Colosseum. Both are meaningful possibilities based on the given ciphertext.

Therefore, the ciphertext alone does not establish one unique meeting location. Additional context or prior information is required to determine Caesar's intended location.

(c) Interesting point about ciphertext-only attacks:

This exercise demonstrates that a ciphertext-only attack can recover possible plaintexts even when the encryption key is unknown.

Since the Caesar cipher has only 26 possible keys, an attacker can test every key and identify meaningful results using language patterns and context.

However, multiple meaningful plaintexts may create ambiguity. Therefore, recovering a meaningful word does not always guarantee that the correct original message has been identified.

3 Problem 2: Affine Cipher Using a Chosen-Plaintext Attack

Problem Statement:

An Affine cipher operates modulo 26. The attacker chooses the plaintext HAHAHA and obtains the ciphertext NONONO. Determine the encryption key and encryption function.

3.1 Cryptanalysis Model

Model Used: Chosen-Plaintext Attack

In a chosen-plaintext attack, the attacker can select a plaintext and obtain its corresponding ciphertext.

By comparing the chosen plaintext and its ciphertext, the attacker can form mathematical equations to determine the encryption key.

3.2 Affine Cipher Formula

The Affine cipher encryption formula is:

$$E(x) = (ax + b) \bmod 26$$

Where:

- a = Multiplicative key.
- b = Additive key.
- x = Plaintext numerical value.

For a valid Affine cipher, the value of a must satisfy:

$$\gcd(a, 26) = 1$$

3.3 (a) Numerical Conversion

Given plaintext:

`HAHAHA`

Given ciphertext:

`NONONO`

Using the alphabet-to-number conversion:

Plaintext	H	A	H	A	H	A
Value	7	0	7	0	7	0

Ciphertext	N	O	N	O	N	O
Value	13	14	13	14	13	14

Thus, the numerical values are:

$$P = (7, 0, 7, 0, 7, 0)$$

$$C = (13, 14, 13, 14, 13, 14)$$

3.4 (b) Forming the Equations Modulo 26

The encryption formula is:

$$E(x) = (ax + b) \bmod 26$$

Using the plaintext letter H and ciphertext letter N:

$$7a + b \equiv 13 \pmod{26}$$

Using the plaintext letter A and ciphertext letter O:

$$0a + b \equiv 14 \pmod{26}$$

Therefore:

$$b \equiv 14 \pmod{26}$$

Substituting $b = 14$ into the first equation:

$$7a + 14 \equiv 13 \pmod{26}$$

$$7a \equiv 13 - 14 \pmod{26}$$

$$7a \equiv -1 \pmod{26}$$

$$7a \equiv 25 \pmod{26}$$

3.5 (c) Determining the Values of a and b

We have:

$$7a \equiv 25 \pmod{26}$$

The multiplicative inverse of 7 modulo 26 is 15 because:

$$7 \times 15 = 105$$

$$105 \bmod 26 = 1$$

Multiplying both sides by 15:

$$a \equiv 25 \times 15 \pmod{26}$$

$$a \equiv 375 \pmod{26}$$

$$a \equiv 375 - 364 \pmod{26}$$

$$\boxed{a = 11}$$

From the second equation:

$$\boxed{b = 14}$$

Checking the validity of the multiplicative key:

$$\gcd(11, 26) = 1$$

Therefore, the key is valid.

3.6 (d) Encryption Function

The Affine cipher encryption function is:

$$E(x) = (ax + b) \bmod 26$$

Substituting $a = 11$ and $b = 14$:

$$\boxed{E(x) = (11x + 14) \bmod 26}$$

3.7 Verification

For H, where $x = 7$:

$$E(7) = (11 \times 7 + 14) \bmod 26$$

$$= (77 + 14) \bmod 26$$

$$= 91 \bmod 26 = 13 = N$$

For A, where $x = 0$:

$$E(0) = (11 \times 0 + 14) \bmod 26$$

$$= 14 = O$$

Thus, the encryption function correctly produces:

$$\boxed{\texttt{HAHAHA} \rightarrow \texttt{NONONO}}$$

4 Problem 3: Affine Cipher Using a Known-Plaintext Attack

Problem Statement:

The ciphertext `CRWWZ` was encrypted using an Affine cipher modulo 26. The attacker knows that the original plaintext begins with `HA`. Determine the encryption and decryption functions and recover the complete plaintext.

4.1 Cryptanalysis Model

Model Used: Known-Plaintext Attack

In a known-plaintext attack, the attacker knows one or more plaintext characters and their corresponding ciphertext characters.

This information is used to form equations and recover the encryption key.

Given:

Known plaintext: HA

Corresponding ciphertext: CR

4.2 (a) Determining the Values of a and b

Using the alphabet-to-number conversion:

$$H = 7, \quad A = 0$$

$$C = 2, \quad R = 17$$

The Affine encryption formula is:

$$E(x) = (ax + b) \bmod 26$$

For the first known pair H $\rightarrow$ C:

$$7a + b \equiv 2 \pmod{26}$$

For the second known pair A $\rightarrow$ R:

$$0a + b \equiv 17 \pmod{26}$$

Therefore:

$$b \equiv 17 \pmod{26}$$

Substituting $b = 17$ into the first equation:

$$7a + 17 \equiv 2 \pmod{26}$$

$$7a \equiv 2 - 17 \pmod{26}$$

$$7a \equiv -15 \pmod{26}$$

$$7a \equiv 11 \pmod{26}$$

The multiplicative inverse of 7 modulo 26 is 15:

$$7 \times 15 = 105$$

$$105 \bmod 26 = 1$$

Multiplying both sides by 15:

$$a \equiv 11 \times 15 \pmod{26}$$

$$a \equiv 165 \pmod{26}$$

$$a \equiv 165 - 156 \pmod{26}$$

$$\boxed{a = 9}$$

Therefore:

$$\boxed{a = 9, \quad b = 17}$$

Checking the validity:

$$\gcd(9, 26) = 1$$

Hence, the key is valid.

4.3 (b) Encryption Function

The general Affine encryption function is:

$$E(x) = (ax + b) \bmod 26$$

Substituting $a = 9$ and $b = 17$:

$$\boxed{E(x) = (9x + 17) \bmod 26}$$

4.4 (c) Finding the Multiplicative Inverse and Decryption Function

The decryption formula for an Affine cipher is:

$$D(y) = a^{-1}(y - b) \bmod 26$$

We need to find the multiplicative inverse of 9 modulo 26.

Testing:

$$9 \times 3 = 27$$

$$27 \bmod 26 = 1$$

Therefore:

$$\boxed{9^{-1} = 3 \pmod{26}}$$

Substituting the values into the decryption formula:

$$D(y) = 3(y - 17) \bmod 26$$

Thus, the decryption function is:

$$\boxed{D(y) = 3(y - 17) \bmod 26}$$

4.5 (d) Decrypting the Complete Ciphertext

Given ciphertext:

`CRWWZ`

The numerical values are:

$$C = 2, \quad R = 17, \quad W = 22, \quad W = 22, \quad Z = 25$$

Using:

$$D(y) = 3(y - 17) \bmod 26$$

For C, where $y = 2$:

$$D(2) = 3(2 - 17) \bmod 26$$

$$= 3(-15) \bmod 26$$

$$= -45 \bmod 26 = 7 = H$$

For R, where $y = 17$:

$$D(17) = 3(17 - 17) \bmod 26$$

$$= 0 = A$$

For W, where $y = 22$:

$$D(22) = 3(22 - 17) \bmod 26$$

$$= 3 \times 5 \bmod 26$$

$$= 15 = P$$

For the second W:

$$D(22) = 15 = P$$

For Z, where $y = 25$:

$$D(25) = 3(25 - 17) \bmod 26$$

$$= 3 \times 8 \bmod 26$$

$$= 24 = Y$$

Therefore, the decrypted plaintext is:

$$\boxed{\texttt{HAPPY}}$$

4.6 Verification

Using the encryption function:

$$E(x) = (9x + 17) \bmod 26$$

For H, $x = 7$:

$$E(7) = (9 \times 7 + 17) \bmod 26$$

$$= 80 \bmod 26 = 2 = C$$

For A, $x = 0$:

$$E(0) = 17 = R$$

For P, $x = 15$:

$$E(15) = (9 \times 15 + 17) \bmod 26$$

$$= 152 \bmod 26 = 22 = W$$

Thus:

$$\boxed{\texttt{HAPPY} \rightarrow \texttt{CRWWZ}}$$

The encryption and decryption results are verified.

5 Problem 4: One-Time Pad Encryption and Decryption

Problem Statement:

Alice uses a One-Time Pad to encrypt the plaintext `RIVER` with the key `NQXTP`. Encrypt the message, decrypt the resulting ciphertext, and explain the requirements of a secure One-Time Pad key.

5.1 Cryptographic Model

Model Used: Symmetric-Key Encryption

The One-Time Pad uses the same secret key for encryption and decryption.

In this exercise, the plaintext and key are provided, so the task is to perform encryption and verify the decryption.

5.2 One-Time Pad Formulas

Encryption formula:

$$C_i = (P_i + K_i) \bmod 26$$

Decryption formula:

$$P_i = (C_i - K_i) \bmod 26$$

Where:

- P_i = Plaintext numerical value.

- K_i = Key numerical value.
- C_i = Ciphertext numerical value.

5.3 (a) Numerical Conversion

Given plaintext:

`RIVER`

Given key:

`NQXTP`

Converting the plaintext into numerical values:

Plaintext	R	I	V	E	R
Value	17	8	21	4	17

Converting the key into numerical values:

Key	N	Q	X	T	P
Value	13	16	23	19	15

Therefore:

$$P = (17, 8, 21, 4, 17)$$

$$K = (13, 16, 23, 19, 15)$$

5.4 (b) Encryption Process

Using the encryption formula:

$$C_i = (P_i + K_i) \bmod 26$$

For the first character:

$$C_1 = (17 + 13) \bmod 26$$

$$= 30 \bmod 26 = 4 = E$$

For the second character:

$$C_2 = (8 + 16) \bmod 26$$

$$= 24 = Y$$

For the third character:

$$C_3 = (21 + 23) \bmod 26$$

$$= 44 \bmod 26 = 18 = S$$

For the fourth character:

$$C_4 = (4 + 19) \bmod 26$$

$$= 23 = X$$

For the fifth character:

$$C_5 = (17 + 15) \bmod 26$$

$$= 32 \bmod 26 = 6 = G$$

Therefore, the ciphertext is:

$$\boxed{\texttt{EYSXG}}$$

5.5 (c) Decryption Process

Using the decryption formula:

$$P_i = (C_i - K_i) \bmod 26$$

The ciphertext is:

$$C = (4, 24, 18, 23, 6)$$

The key is:

$$K = (13, 16, 23, 19, 15)$$

For the first character:

$$P_1 = (4 - 13) \bmod 26$$

$$= -9 \bmod 26 = 17 = R$$

For the second character:

$$P_2 = (24 - 16) \bmod 26$$

$$= 8 = I$$

For the third character:

$$P_3 = (18 - 23) \bmod 26$$

$$= -5 \bmod 26 = 21 = V$$

For the fourth character:

$$P_4 = (23 - 19) \bmod 26$$

$$= 4 = E$$

For the fifth character:

$$P_5 = (6 - 15) \bmod 26$$

$$= -9 \bmod 26 = 17 = R$$

Therefore, the recovered plaintext is:

$$\boxed{\texttt{RIVER}}$$

The decrypted message matches the original plaintext.

5.6 (d) Requirements of a Secure One-Time Pad

A One-Time Pad provides perfect secrecy only when its key satisfies the following requirements:

1. **Randomness:**

   The key must be generated using truly random values. Predictable keys can make the encrypted message vulnerable to attacks.

2. **Equal Length:**

   The key must have the same length as the plaintext. Every plaintext character must have a corresponding key character.

3. **Used Only Once:**

   The key must never be reused for another message. Reusing a key can reveal relationships between different plaintext messages.

4. **Secure Key Distribution:**

   The key must be shared securely between the sender and receiver. If an attacker obtains the key, the encrypted message can be decrypted.

Therefore, a One-Time Pad is theoretically unbreakable when the key is truly random, kept secret, equal in length to the message, and used only once.

6 Conclusion

Through this assignment, different classical cryptographic techniques and cryptanalysis models were studied and implemented using modular arithmetic.

The Caesar cipher exercise demonstrated how a ciphertext-only attack can be performed by testing all possible keys.

The Affine cipher exercises demonstrated how chosen-plaintext and known-plaintext attacks can be used to determine encryption keys through modular equations.

The One-Time Pad exercise demonstrated the encryption and decryption process using a secret key and highlighted the importance of randomness, key length, secrecy, and single use.

Overall, these exercises provided practical understanding of classical encryption algorithms, modular arithmetic, and basic cryptanalysis techniques.

7 Reference

1. Cryptology Assignment Repository – GitHub.

   `https://github.com/Shreyas-2004-ai/Cryptology-`

"Success is not obtained overnight. It comes in installments; you get a little bit today, a little bit tomorrow until the whole package is given out. The day you procrastinate, you lose that day's success."

— Israelmore Ayivor